

LogAnalyzer Test Cases

Command name: loganalyzer

Compile command:

gcc loganalyzer.c -o loganalyzer

Test Case 1: Normal log file

File:

sample.log

Command:

./loganalyzer -f sample.log

Expected Output:

Log Analysis Result :

File name : sample.log

Total lines : 8

ERROR count : 2

WARNING count : 2

INFO count : 4

Purpose:

This test checks that the program correctly counts total lines and log levels.

Test Case 2: Missing argument

Command:

./loganalyzer

Expected Output:

Usage: ./loganalyzer -f <logfile>

Purpose:

This test checks getopt() argument handling.

Test Case 3: File does not exist

Command:

./loganalyzer -f missing.log

Expected Output:

Error opening file: No such file or directory

Purpose:

This test checks error handling when the input file is invalid.

Test Case 4: Empty file

Create file:

```
touch empty.log
```

Command:

```
./logalyzer -f empty.log
```

Expected Output:

The file is empty.

Purpose:

This test checks if the program handles empty files safely.